# Smart Campus Emergency Logistics Optimization

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# Chapter 1

# Introduction

Managing the distribution of essential supplies is an important task in large environments such as university campuses. A campus contains many buildings including academic blocks, laboratories, event spaces, and service facilities. Each of these locations may require regular supplies such as food, medical items, or other essential materials.

In emergency situations, delivering these resources quickly and efficiently becomes even more important. However, planning such logistics manually can be difficult because many factors must be considered. These include the location of warehouses, the demand of different facilities, and the cost of transporting resources.

This project explores how optimization techniques can help design a better logistics system for a campus environment. By using a mathematical model, the system can identify the best warehouse locations and determine how supplies should be distributed to meet demand while keeping costs as low as possible.

The goal of this work is to demonstrate how computational tools and optimization methods can improve logistics planning and support smarter decision making.

# Chapter 2

# Problem Statement

A university campus contains multiple buildings that require a steady supply of essential resources. Delivering these resources efficiently becomes challenging when several facilities must be served from a limited number of warehouse locations.

If logistics decisions are made manually, it may lead to problems such as higher transportation cost, inefficient allocation of supplies, and delays in delivery. In addition, relying on a single warehouse may create risk if that warehouse becomes unavailable.

The objective of this project is to design an optimized supply distribution network that can support campus facilities in an efficient and reliable way.

The proposed system aims to:

- Reduce the total cost of logistics operations
- Ensure that all campus facilities receive sufficient supplies
- Respect the capacity limits of warehouses
- Maintain reliability by selecting multiple warehouses

# Chapter 3

# Dataset Description

To build the optimization model, a dataset representing campus facilities and potential warehouse locations was created.

## 3.1 Campus Facilities

The facilities considered in the model include important academic blocks, event venues, and service areas within the campus. Each facility is assigned a daily demand value which represents the amount of resources required.

| Facility ID | Facility Name | Daily Demand |
|---|---|---|
| RB_BLOCK | RB Block | 45 |
| VB_BLOCK | VB Block | 40 |
| CLC_BLOCK | CLC Block | 35 |
| AB_BLOCK | AB Block | 42 |
| NORTH_BLOCK | North Block | 38 |
| SOUTH_BLOCK | South Block | 36 |
| AMENITY_BLOCK | Amenity Block | 50 |
| AK_BLOCK | AK Block | 37 |
| MINI_AUDITORIUM | Mini Auditorium | 30 |
| AMPHITHEATRE | Amphitheatre | 28 |
| CENTRAL_COMPLEX | Central Complex | 55 |
| LAB_BLOCKS | Engineering Labs | 48 |
| MAIN_ENTRY | Main Entry | 25 |

Table 3.1: Campus Facility Demand

## 3.2 Warehouse Locations

Four possible warehouse locations were considered. Each warehouse has a defined storage capacity and operational cost.

| Warehouse ID | Capacity | Construction Cost | Operational Cost/day |
|---|---|---|---|
| WH_NORTH | 420 | 310000 | 850 |
| WH_SOUTH | 380 | 295000 | 780 |
| WH_EAST | 450 | 330000 | 910 |
| WH_WEST | 400 | 305000 | 820 |

Table 3.2: Warehouse Information

# Chapter 4

# Mathematical Model

The logistics planning problem is formulated as an optimization problem. The model determines which warehouses should be activated and how resources should be distributed to each facility.

## 4.1 Decision Variables

Two types of decision variables are used in the model:

- $x_{wfr}$ represents the quantity of resource transported from warehouse $w$ to facility $f$.
- $y_w$ is a binary variable indicating whether a warehouse is selected.

## 4.2 Objective Function

The objective is to minimize the total logistics cost including construction cost, operational cost, and transportation cost.

$$\min \sum_{w} \left( \frac{C_w}{8} + 365 O_w \right) y_w + \sum_{w,f,r} d_{wf} x_{wfr} \tag{4.1}$$

## 4.3 Constraints

The model must satisfy several constraints.

**Demand Constraint**

$$\sum_{w} x_{wfr} \geq D_f \tag{4.2}$$

**Capacity Constraint**

$$\sum_{f} x_{wfr} \leq Cap_w \cdot y_w \tag{4.3}$$

**Redundancy Constraint**

$$\sum_{w} y_w \geq 2 \tag{4.4}$$

These constraints ensure that demand is satisfied, warehouses are not overloaded, and at least two warehouses remain active.

# Chapter 5

# Implementation

The optimization model was implemented using Python. Several libraries were used to support different parts of the system.

- Python programming language
- Pandas for data processing
- PuLP for optimization modelling
- Folium for map visualization

Distances between warehouses and facilities were calculated using geographic coordinates. The model was then solved using the CBC solver included in the PuLP library.

The complete source code for the project is available at:
`https://github.com/Fahadbinhashim2005/Micro-project`

# Chapter 6

# Results

After running the optimization model, the system identified the most efficient warehouse configuration.

## 6.1 Selected Warehouses

The optimization results indicate that the following warehouses provide the most efficient distribution strategy:

- WH_SOUTH
- WH_WEST

## 6.2 Cost of Operation

The optimal annual logistics cost obtained from the model is approximately:

**$2.53 Million**

## 6.3 Visualization of Logistics Network

The final distribution network is visualized using a map that shows warehouse locations, campus facilities, and supply routes.

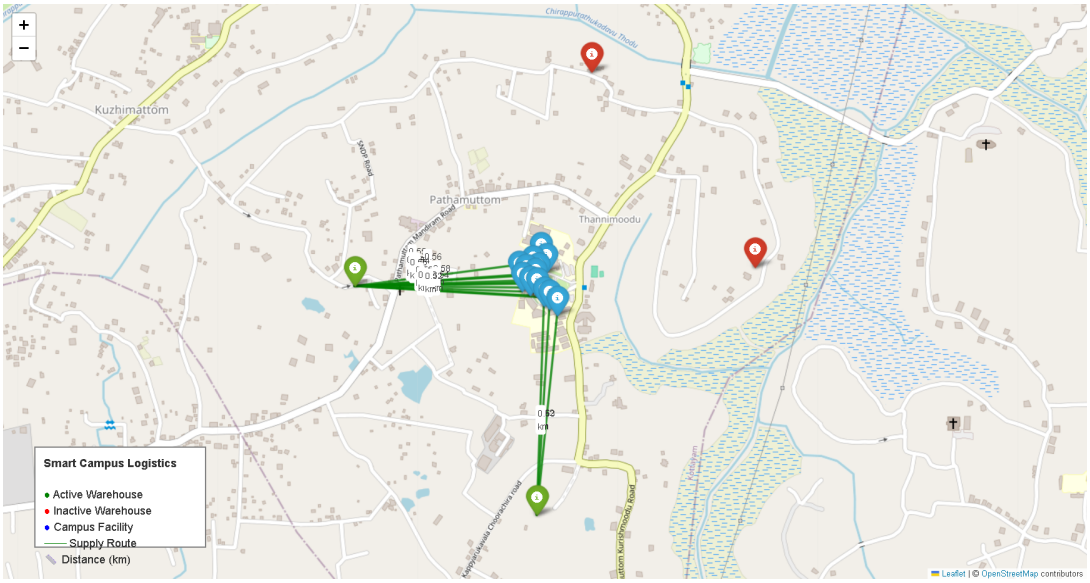


Figure 6.1: Optimized Campus Logistics Network

# Chapter 7

# Conclusion

This project demonstrates how optimization techniques can be used to design an efficient logistics system for a campus environment.

By modelling the problem mathematically, the system is able to determine the best warehouse locations and distribution plan while minimizing overall cost.

The results show that computational approaches can help improve logistics planning and support better decision making, especially during emergency situations.

Future improvements may include incorporating real-time data such as traffic conditions and dynamic demand changes.

# Chapter 8

# References


- PuLP Optimization Library Documentation
- Python Data Analysis Library (Pandas)